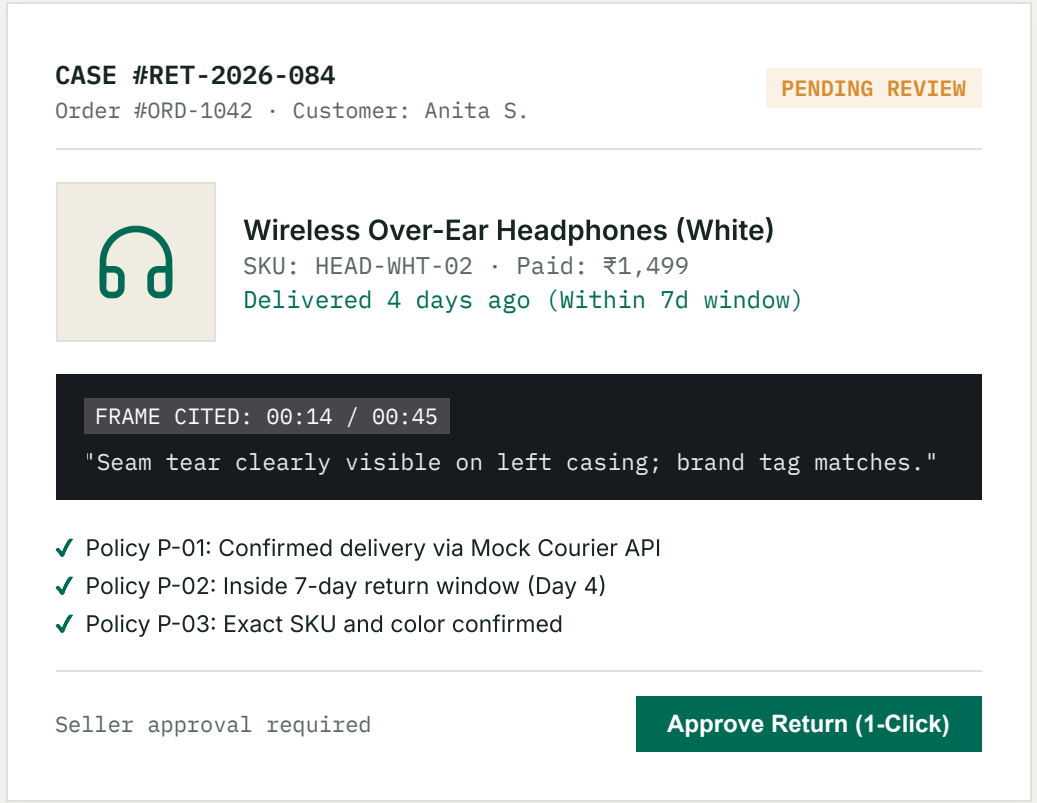
ReturnLens

Evidence-backed return triage for small online sellers.

Verifying buyer unboxing videos against mock store truth before any return label or refund draft is prepared for one-click seller sign-off.

Team Rookie | 3 Members | 24-Hour Prototype | Mock Data & Staged Evidence



THE PROBLEM · SOLO SELLER BOTTLENECK

The return decision is made from memory, before the evidence is checked.

TAB 1: BUYER CHAT

"Received damaged item. Need urgent refund."
unboxing_clip_34s.mp4

TAB 2: STORE ADMIN

ORD-8492 · ₹1,499
SKU: HEAD-WHT-02
Customer: Rohit K.

TAB 3: SHIPPING PORTAL

Delivered: 26 Oct (4d ago)
Tracking: TRK-98214-DEL
Carrier: MockPost

TAB 4: POLICY PDF

"7-day limit from delivery.
Tags intact. Original box.
Refund cap: ₹4,999."

A Wrong Decision Costs Both Ways

- **Unjust Approval:** Loses money, shipping costs, and inventory to wardrobing or swapped items.
- **Unfair Denial:** Destroys customer goodwill, loses repeat buyers, and invites harsh marketplace disputes.

Sourced Industry Benchmark

Fraud estimates differ by definition: **9%** (NRF / Happy Returns 2025, fraud only) vs **15.14%** (Appriss / Deloitte 2024, fraudulent & abusive claims rate).

*U.S. retail estimates; not measured on our target sellers.

The gap is not a missing returns portal. It is that no single record ties order data, video frame, policy clause, and decision together.

SOLUTION ARCHITECTURE · END-TO-END WORKFLOW

Verify the video first. Act only after the seller approves.

01

Receive Request

Buyer submits claim via mock web chat with 30–60s video proof.

`video + order_id + claim`

02

Retrieve Data

Calls mock store & shipping APIs for SKU, price, and delivery date.

`mock order & tracking APIs`

03

Build Evidence

Extracts video frames via CPU ffmpeg (~1 fps) and anchors claims.

`server frame manifest`

04

Check Policy

Code evaluates policy clauses; critic checks cited frames.

`P-01 to P-08 code checks`

05

Seller Approval

Seller clicks once to issue mock return label and refund draft.

`label + draft pending`

↳ **Fallback Branch: Request specific evidence → resume same case**

If video is blurry or missing tag, agent prompts for a specific photo. Never auto-denies; state is preserved in the same case session.

MOCK RETURN LABEL

CARRIER: MOCK-EXPRESS · #RL-9021

Status: pending_return_receipt



ARCHITECTURE & SAFETY · DETERMINISTIC CONTROL

Policy checks and seller approval control actions

01 · OBSERVE

Models Inspect Evidence

Models process buyer messages and server-extracted video frames.

- Outputs structured visual observations
- Identifies damage timestamps and tags
- *Constraint:* Models observe, but observations can be imperfect and are never final verdicts

02 · CHECK

Deterministic Policy & Critic

Pure code evaluates server-enforced rules; LLMs cannot override.

- **Code Policy Engine:** Delivery date, 7-day window, exact SKU match, refund cap calculation
- **Independent Critic:** Reviews cited frames and flags unsupported observations
- Fail-closed rule: Discrepancy triggers escalation

03 · AUTHORIZE

Mandatory Seller Gate

Human-in-the-loop sign-off required for all irreversible steps.

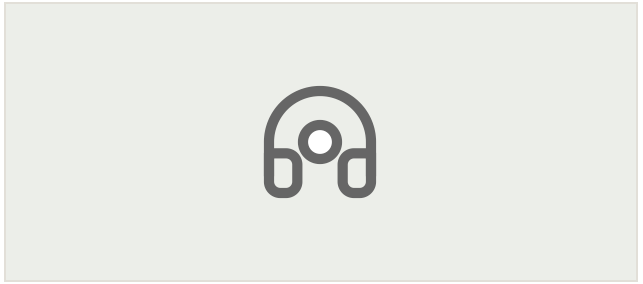
- No automated refunds or label generation
- Presents 10-second proof package
- Draft created in `pending_return_receipt`
- Immutable JSONL audit trail of all steps

DECISION FUNCTION: Contradiction → Escalate | Hard Fail → Escalate | Unclear Evidence → Ask for Photo | All Pass → Seller Review

EVIDENCE BENCHMARK · VISUAL AUDIT PROOF

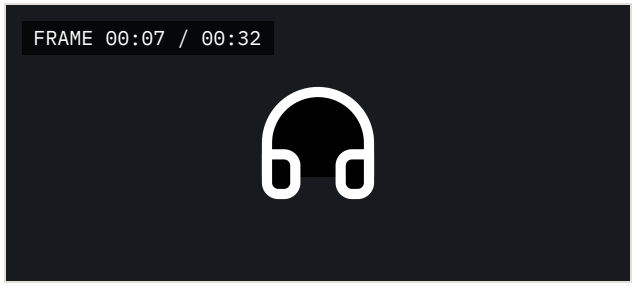
Case B: A mismatch that needs review

STORE API RECORD — ORDER TRUTH



SKU: HEAD-WHT-02
Product: Wireless Headset
Color: White · ₹1,499

VIDEO FRAME [00:07] — CLAIM EVIDENCE



Observed: HEAD-BLK-09
Color: Black
Status: Brand tag missing

Policy P-03 Check: FAIL (MISMATCH)

NEEDS SELLER REVIEW · ESCALATED

No automatic denial. No return label or refund draft issued at this stage.

Seller Resolution Options:

- Review possible warehouse mis-shipment
- Request outside-box shipping label photo
- Hold for physical inspection upon return

Case A · Ready:
ORD-1042 (₹1,499) match → 1-click label & draft

Case B · Mismatch:
Different color/SKU → Escalated for review

Case C · Unclear:
Dark video (br: 28) → Ask photo, resume case

Case D · Context:
Continuous take showing box label to item

DELIVERY PLAN · TEAM ROOKIE

Buildable in 24 hours by 3 people, if the scope stays narrow.

00h – 06h	**Mock APIs & Core Return Flow**	FastAPI mock store & shipping services, SQLite seed data, core LangGraph state graph.
06h – 12h	**Evidence Pipeline & Policy Checks**	ffmpeg CPU frame extraction (~1 fps), multimodal prompt grounding, deterministic code policy rules.
12h – 18h	**Integration & 4 Demo Cases**	End-to-end testing of Cases A, B, C, and D, independent critic validation, fallback error handling.
18h – 24h	**Demo Recording & Submission**	Seller review console polish, 3-minute video walk-through, GitHub README, submission package.

Person A (Backend & Logic)
Backend, agent graph, deterministic policy engine, critic, audit log.

Person B (Multimodal & UI)
Video pipeline, frame indexer, seller console, evidence pack view.

Person C (QA & Integration)
Integration, mock store data, 4 demo cases, fallback testing, demo video.

Strict 24-Hour Scope Guardrails:

- Cut first: WhatsApp direct, multi-seller, audio, real payment gateway.
- Never cut: Code policy engine, critic check, seller gate, 4 demo cases.
- Cap: `0 < amount ≤ min(₹4,999, remaining, paid)`